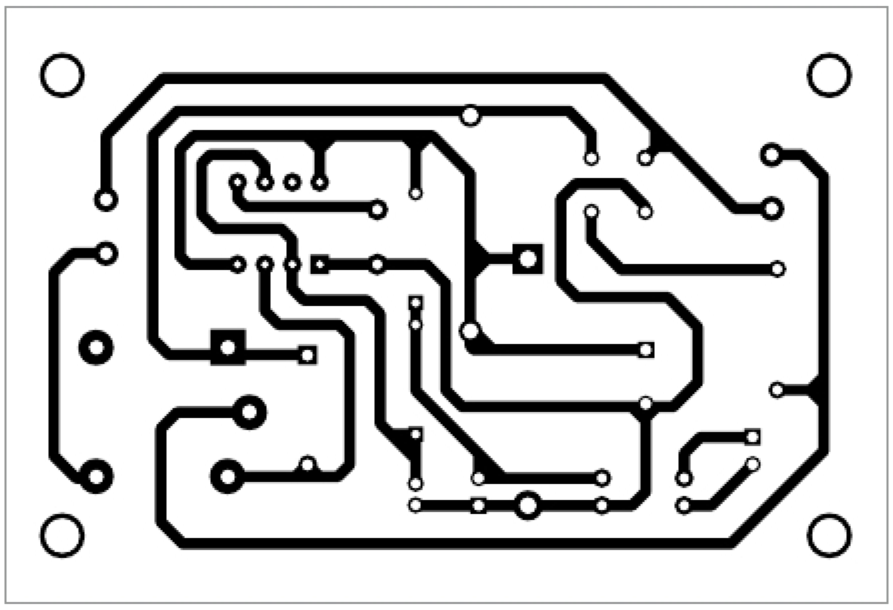

Fig. 3: Actual-size PCB layout for the smart bulb holder

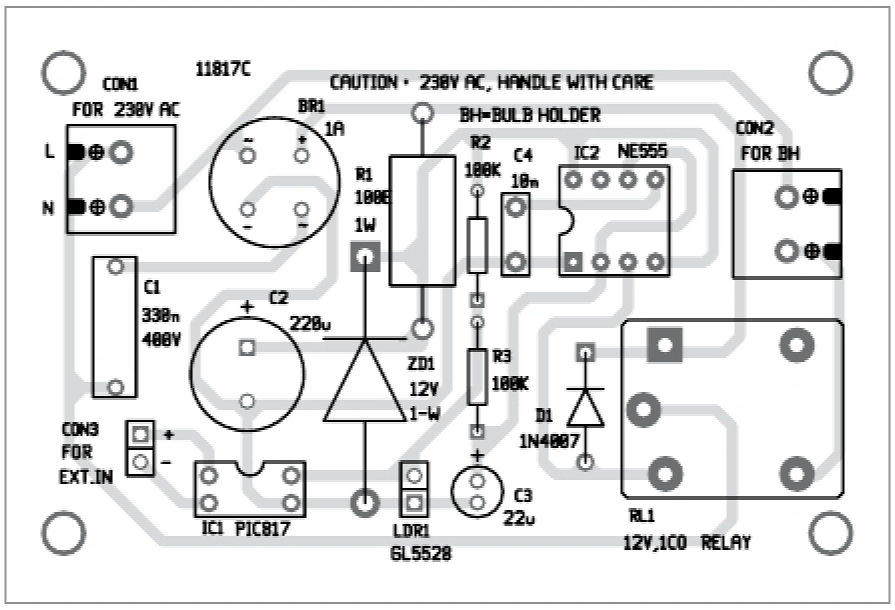

Fig. 4: Components layout of the PCB

PARTS LIST

Semiconductors:	
IC1	- PC817 opto-coupler
IC2	- NE555 timer
BR1	- 1A bridge rectifier
D1	- 1N4007 rectifier diode
ZD1	- 12V, 1-watt Zener diode
Resistors (all 1/4-watt, ±5% carbon), unless stated otherwise:	
R1	- 100-ohm, 1-watt
R2, R3	- 100-kilo-ohm
Capacitors:	
C1	- 330nF, 400V ceramic disk
C2	- 220µF, 25V electrolytic
C3	- 22µF, 25V electrolytic
C4	- 10nF ceramic disk
Miscellaneous:	
CON1, CON2	- 2-pin terminal connector
CON3	- 2-pin connector
LDR1	- GL5528 LDR
RL1	- 12V, single changeover relay
	- Bulb holder

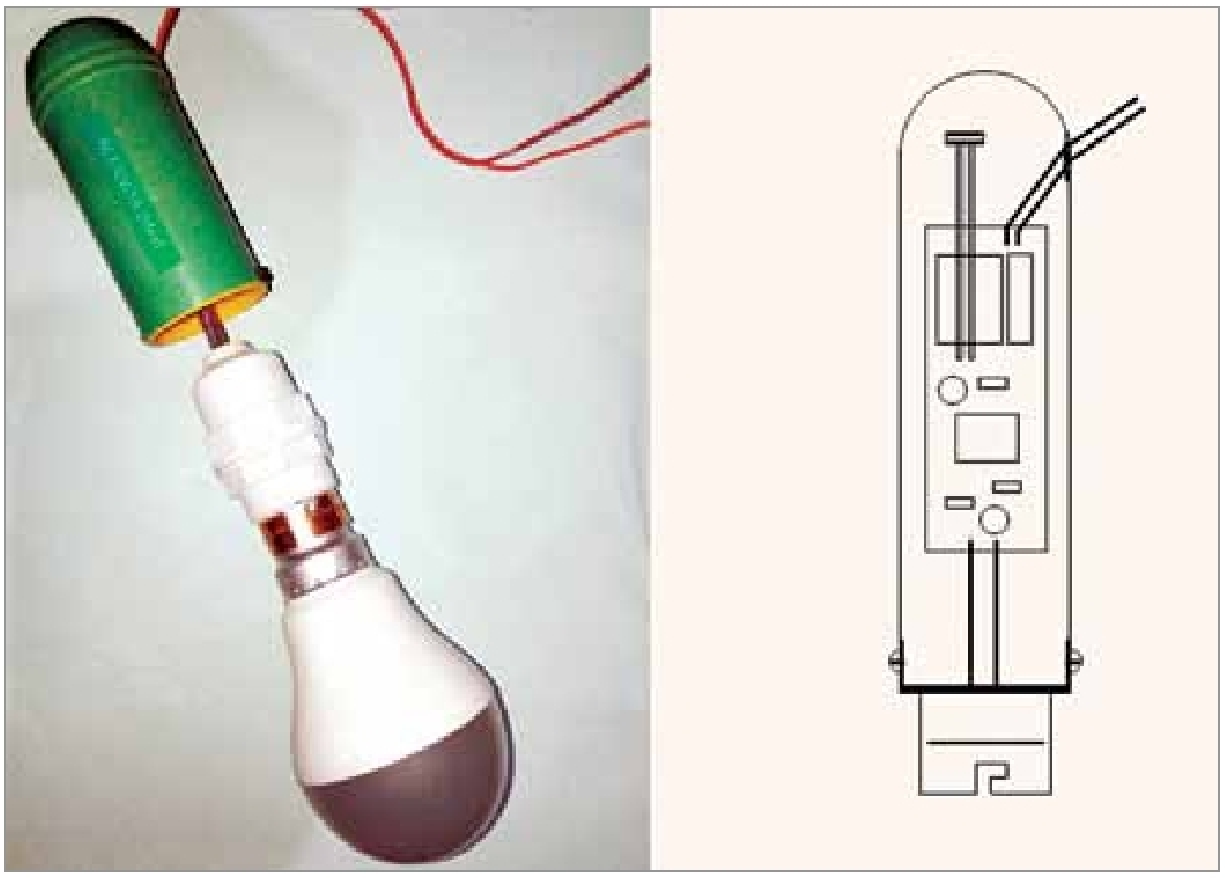

Fig. 5: A sample enclosure layout

As you might notice in the diagram, there's also a galvanically-isolated switch/sensor input in the device built around a common photocoupler PC817 (IC1). The option allows you to remove LDR1 and control the smart bulb holder from an external IoT (or similar) gadget if you desire to have a simple home automation project. It is to be noted here that if the photocoupler is in active state, the bulb will switch off. Needless to say, the photocoupler requires a current-limiter resistor at its input to set the typical forward current close to 20mA at 1.2V typical forward voltage.

Construction and testing

The entire circuit can be assembled on a rectangular piece of perforated or general-purpose PCB (refer author's prototype). An actual-size PCB layout for the smart bulb holder is, however, shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect the 230V AC mains across CON1.

A small (waterproof) translucent enclosure is a must for this project. Prototype enclosures come in many shapes and sizes. It would be better to make your own enclosure if you've access to a 3D printer as it'll give your model a polished look. A sample enclosure layout is also depicted here in Fig. 5 for your quick reference.

Author's note. At first, the dimmable daylight LED bulb enclosed prototype was used for performance evaluation. Later, it was mounted on a garden lamp pole, and the real-world results (till now) are very promising. This smart bulb holder not only works as an energy-saving outdoor light bulb but also serves as a security light bulb or home occupancy indicator to deter prowlers even when you're on a vacation.

Caution. This circuit involves the direct use of 230V AC mains voltage, which is dangerous, and the chances of getting an electric shock are high if not handled properly. If you are not used to handling such voltages, please get the help of a qualified person. EFY

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